

Experimental Demonstration of 3D Optical Wireless Positioning with Single Beam-Steered LED and Empirical Radiation Pattern

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Abstract—TBD

Index Terms—TBD

I. INTRODUCTION

II. WORKING PRINCIPLES OF 3D OWP

A. System Model

B. Direction Finding

C. Distance Recovery

Se consideraría dos metodos (a) el PD apunta hacia el LED asumiendo que el receptor tiene capacidad de reorientación, se apuntan entre ellos y se recupera la distancia, en total $K+1$ reorientaciones (b) el PD se mantiene fijo y se estima la distancia a partir de la información previamente registrada de las K orientaciones.

D. Transmitter Irradiation Pattern and Positioning Estimation Enhancement

(a) Se discute acerca del modelo de irradiancia del LED transmisor ideal Lambertiano versus el patron de irradiacion experimental. (b) Se propone un método de como usar la información experimental del patron de irradiancia para mejorar la estimación de posición.

III. DETAILS ON THE PROTOTYPE IMPLEMENTATION

A. 3D Robotic Testbed Configuration

Entre las potenciales modificaciones experimentales a discutir se encuentran:

- Mejora de disipacion de calor en el LED transmisor. Se propone añadir un disipador de calor de aluminio montado en el Gimbal en conjunto con la compra de un LED montado en un star PCB. Esto permitiría elevar la corriente a la corriente nominal del LED de 1A (en el GLOBECOM paper se uso 750mA).
- Actualmente la limitante para tener un testbed de 3m x 3m x 1.4m no está en el sistema de transmisión SBL, sino en los limites del FOV del PD. El PD que usamos es el BPX 61 con un FOV/2 de $\approx 60^\circ$, lo que geométricamente limita el tamaño del testbed a 2.4 x 2.4 x 1.2 m, fuera del mismo el RSS recibido cae drasticamente por el

angulo de recepcion, incluso con un perfecto LOS o beam-steering el RSS la informacion no es suficiente para determinar la posicion. La solución seria elaborar la PCB con el PD S6967 de un FOV/2 mayor ($\approx 90^\circ$), con esto se podría llegar a 3m x 3m x 1.4m.

- Aumento en las posiciones evaluadas. Actualmente en el paper GLOBECOM 2026 se evaluaron 300 posiciones. Manteniendo el grid de 0.2m : En el caso de emplear el BPX 61 llegaríamos a 845 posiciones ($h = 0.4m$ a $1.4m$, $w = 2.4m$, $l = 2.4m$), a 1014 ($h = 0.2m$ a $1.4m$, $w = 2.4m$, $l = 2.4m$); mientras que con el S6967 llegaríamos a 1280 ($h = 0.4m$ a $1.4m$, $w = 3m$, $l = 3m$) o 1536 ($h = 0.2m$ a $1.4m$, $w = 3m$, $l = 3m$). En todos los casos superamos las 300 de muestras del GLOBECOM.
- (a) del lado del transmisor: En la descripción del mecanismo gimbal añadir las ecuaciones de rotación y transformación de coordenadas. (b) del lado del receptor: Describir las ecuaciones que transforman la posición deseada del PD en el sistema de referencia del testbed en el vector de posicion y orientación en el sistema de coordenadas del robot.

B. Experimental Protocol for the SBL data acquisition

- Considerar registrar en la misma posición diferentes ángulos del PD para evaluar la robustez ante el tilt en PD del método SBL en 3D o especialmente en finding-Direction.

C. Experimental Protocol for the empirical irradiation pattern

- Se presenta el protocolo para la adquisicion del patron de irradiación del LED.

IV. EVALUATION OF THE DIRECTION FINDING PERFORMANCE

V. EVALUATION OF THE POSITIONING PERFORMANCE

VI. EVALUATION OF THE EMPIRICAL IRRADIATION PATTERN FOR IMPROVING DIRECTION AND POSITION ESTIMATION

VII. CONCLUSION AND FUTURE WORKS

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